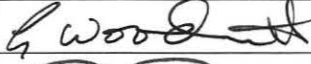
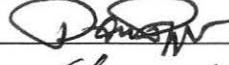


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TITLE: GXP and Non-GXP Study Report Review and Approval Form				
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☐ Vendor Study Protocol		☐ Vendor Study Report		☒ Lpath Study Report	
Lpath Study No.: LT3114-PHA-0010-R		Version No.: 01		☒ Non GXP ☐ GCP	☐ GLP ☐ GMP
Product Name: LT3114		Test Article: LT3114; LT3211; LT3212; LT3611			
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Function	Name (Print)	Signature	Date
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Executive Summary

This pharmacokinetic (PK) study in Sprague-Dawley rats was conducted at Biotox Sciences (San Diego, CA) and Lpath (San Diego, CA). The purpose of this study was to establish the systemic concentrations of humanized anti-lysophosphatidic acid (LPA) antibodies in the rat dosed by an intravenous (IV) route of administration. Measuring these concentrations will allow calculation of several pharmacokinetic parameters useful in further development of these antibodies.

Cannulated male Sprague Dawley rats received a 10 mg/kg IV bolus dose of a single humanized anti-LPA antibody (LT3114, LT3211, LT3212 or LT3611). Serial blood samples were collected from an implanted femoral vein cannula for up to 21 days post dosing for determination of anti-LPA antibody plasma concentrations. Plasma concentrations of humanized anti-LPA antibody were quantified using a direct-binding ELISA. PK data analysis using noncompartmental methods (software package WinNonlin version 5.3 (Pharsight, Inc.; Mountain View, CA) was conducted.

Following a single IV dose of LT3114 to male Sprague Dawley rats, LT3114 plasma concentrations were measurable for up to 20 days post dosing. LT3114 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to 2 days after administration of a 10 mg/kg IV bolus. LT3114 plasma concentrations declined about three-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3114 plasma concentrations declined very slowly with an approximate terminal half-life of 3 weeks or longer. Mean LT3114 plasma concentrations at 30 minutes were 178 µg/mL and the mean AUC_{0-last} was 22.9 mg.h/mL. A significant portion of the AUC occurred beyond 20 days.

LT3211 plasma concentrations were measurable for up to 6 days post IV bolus dosing of 10 mg/kg LT3211. LT3211 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to one to two days after administration of a 10 mg/kg IV bolus. LT3211 plasma concentrations declined about seven to ten-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3211 plasma concentrations declined to non-measurable concentrations within two weeks post IV bolus dosing with an approximate terminal half-life of 5 days. Mean LT3211 plasma concentrations at 30 minutes were 181 µg/mL and the mean AUC_{0-last} was 3.94 mg.h/mL.

For the 10 mg/kg IV bolus dose of LT3212, plasma concentrations were measurable for up to 3 days post dosing. LT3212 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to one day after administration of a 10 mg/kg IV bolus. LT3212 plasma concentrations declined about five to seven-fold from 30 minutes to one day post IV bolus dosing. Thereafter, LT3212 plasma concentrations declined to non-measurable concentrations within 4 days post IV bolus dosing with an approximate terminal half-life of 1

STUDY REPORT: LT3114-PHA-010-R	
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day. Mean LT3212 plasma concentrations at 30 minutes were 145 µg/mL and the mean AUC_{0-last} was 2.29 mg.h/mL.

Following a 10 mg/kg IV bolus dose of LT3611, plasma concentrations were measurable for up to 9 days post dosing. LT3611 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to two days after administration of a 10 mg/kg IV bolus. LT3611 plasma concentrations declined about six to eight-fold from 30 minutes to two day posts IV bolus dosing. Thereafter, LT3611 plasma concentrations declined to non-measurable concentrations within 2 weeks post IV bolus dosing with an approximate terminal half-life of 3 days. Mean LT3611 plasma concentrations at 30 minutes were 194 µg/mL and the mean AUC_{0-last} value was 6.57 mg.h/mL.

Monoclonal antibody plasma concentrations were similar for all four monoclonal antibodies at 30 minutes following a 10 mg/kg IV bolus. Of the four anti-LPA monoclonal antibodies, LT3114 plasma concentrations were measurable throughout the 20 day study period. LT3114 had an estimated terminal half-life of 3 weeks or longer whereas the mean estimated terminal half-life values for the other three monoclonal antibodies (e.g., LT3211, LT3212 and LT3611) were all less than one week.

In summary, noncompartmental PK parameters were generated for humanized anti-LPA antibodies in a Sprague Dawley rat after a single IV bolus administration of 10 mg/kg.

**Pharmacokinetic Evaluation of Humanized Anti-LPA Antibodies (Abs) in the Sprague
Dawley rat**

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ABBREVIATIONS

Abs	Antibodies
AUC	Area under the curve
BQL	Below the limit of quantification of the assay
C ₀	Extrapolated plasma concentration at time zero after an IV bolus administration
CL	Systemic clearance
C _{last}	Last measurable concentration
C _{max}	Maximum plasma concentration
ELISA	Enzyme linked immunosorbent assay
F	Absolute bioavailability
GLP	Good laboratory practices
h	Hour
IV	Intravenous
kg	Kilogram
λ _z	Lambda-z; terminal phase rate constant, slope of the terminal phase
LPA	Lysophosphatidic acid
LT3114, LT3211, LT3212, LT3611	Humanized anti-lysophosphatidic acid (LPA) monoclonal antibody
mAb	Monoclonal antibody
µg	Microgram
mg	Milligram
mL	Milliliter
MRT	Mean residence time
NA	Not applicable
NC	Not calculated
NR	Not reported
ng	Nanogram
%CV	Percent coefficient of variation
PK	Pharmacokinetic(s)
rpm	Revolutions per minute
SD	Standard deviation
T _{last}	Time of the last measurable concentration

$T_{\max}$	Time of maximum concentration
V_c	Volume of distribution of central compartment
V_{ss}	Volume of distribution at steady state
V_z	Apparent volume of distribution during elimination phase

1 SUMMARY

This pharmacokinetic (PK) study in Sprague-Dawley rats was conducted at Biotox Sciences (San Diego, CA) and Lpath (San Diego, CA). The purpose of this study was to establish the systemic concentrations of humanized anti-lysophosphatidic acid (LPA) antibodies in the rat dosed by an intravenous (IV) route of administration. Measuring these concentrations will allow calculation of several pharmacokinetic parameters useful in further development of these antibodies.

Cannulated male Sprague Dawley rats received a 10 mg/kg IV bolus dose of a single humanized anti-LPA antibody (LT3114, LT3211, LT3212 or LT3611). Serial blood samples were collected from an implanted femoral vein cannula for up to 21 days post dosing for determination of anti-LPA antibody plasma concentrations. Plasma concentrations of humanized anti-LPA antibody were quantified using a direct-binding ELISA. PK data analysis using noncompartmental methods (software package WinNonlin version 5.3 (Pharsight, Inc.; Mountain View, CA) was conducted.

Following a single IV dose of LT3114 to male Sprague Dawley rats, LT3114 plasma concentrations were measurable for up to 20 days post dosing. LT3114 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to 2 days after administration of a 10 mg/kg IV bolus. LT3114 plasma concentrations declined about three-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3114 plasma concentrations declined very slowly with an approximate terminal half-life of 3 weeks or longer. Mean LT3114 plasma concentrations at 30 minutes were 178 µg/mL and the mean AUC_{0-last} was 22.9 mg.h/mL. A significant portion of the AUC occurred beyond 20 days.

LT3211 plasma concentrations were measurable for up to 6 days post IV bolus dosing of 10 mg/kg LT3211. LT3211 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to one to two days after administration of a 10 mg/kg IV bolus. LT3211 plasma concentrations declined about seven to ten-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3211 plasma concentrations declined to non-measurable concentrations within two weeks post IV bolus dosing with an approximate terminal half-life of 5 days. Mean LT3211 plasma concentrations at 30 minutes were 181 µg/mL and the mean AUC_{0-last} was 3.94 mg.h/mL.

For the 10 mg/kg IV bolus dose of LT3212, plasma concentrations were measurable for up to 3 days post dosing. LT3212 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to one day after administration of a 10 mg/kg IV bolus. LT3212 plasma concentrations declined about five to seven-fold from 30 minutes to one day post IV bolus dosing. Thereafter, LT3212 plasma concentrations declined to non-measurable concentrations within 4 days post IV bolus dosing with an approximate terminal half-life of 1 day. Mean LT3212 plasma concentrations at 30 minutes were 145 µg/mL and the mean AUC_{0-last} was 2.29 mg.h/mL.

Following a 10 mg/kg IV bolus dose of LT3611, plasma concentrations were measurable for up to 9 days post dosing. LT3611 concentrations declined in a bi-exponential fashion with the initial phase lasting for up to two days after administration of a 10 mg/kg IV bolus. LT3611 plasma concentrations declined about six to eight-fold from 30 minutes to two day posts IV bolus dosing. Thereafter, LT3611 plasma concentrations declined to non-measurable concentrations within 2 weeks post IV bolus dosing with an approximate terminal half-life of 3 days. Mean LT3611 plasma concentrations at 30 minutes were 194 µg/mL and the mean AUC_{0-last} value was 6.57 mg.h/mL.

Monoclonal antibody plasma concentrations were similar for all four monoclonal antibodies at 30 minutes following a 10 mg/kg IV bolus. Of the four anti-LPA monoclonal antibodies, LT3114 plasma concentrations were measurable throughout the 20 day study period. LT3114 had an estimated terminal half-life of 3 weeks or longer whereas the mean estimated terminal half-life values for the other three monoclonal antibodies (e.g., LT3211, LT3212 and LT3611) were all less than one week.

In summary, noncompartmental PK parameters were generated for humanized anti-LPA antibodies in a Sprague Dawley rat after a single IV bolus administration of 10 mg/kg.

2 INTRODUCTION

This study was conducted to provide initial noncompartmental pharmacokinetic (PK) parameters after intravenous (IV) bolus administration of four humanized anti-lysophosphatidic acid (LPA) antibodies in the Sprague Dawley rat.

3 MATERIALS AND METHODS

3.1 Study Design

Healthy, male, cannulated Sprague Dawley rats were assigned to study groups as shown in Table 1. Rats weighed between 240 to 260 grams and were approximately 6 to 8 weeks of age. Rats were acclimatized for 2 days prior to the study. A total of 12 rats participated in the study; 3 rats per each group. An IV bolus dose of 10 mg/kg of one of four humanized anti-LPA antibodies (LT3114, LT3211, LT3212 or LT3611) were administered to the 3 male rats in each group. Serial blood samples were taken post IV bolus dosing at the times specified in Table 1.

Table 1. Study design

Antibody	Dose	Route	Animals/ Number of Groups	Time of blood collection post IV dosing
LT 3114, LT 3211, LT 3212, LT 3611	10mg/kg	IV bolus	3/4	30 min
			3/4	3 hours
			3/4	6 hours
			3/4	9 hours
			3/4	12 hours
			3/4	26 hours
			3/4	32 hours
			3/4	48 hours
			3/4	3 days
			3/4	4 days
			3/4	6 days
			3/4	14 days
			3/4	20 days
Total rats:			12	
Antibody amount (mg)			10 mg per group	

3.2 Pharmacokinetic Sample Collection and Analysis

Serial blood samples were obtained from 3 male rats per group via an implanted femoral vein cannula. The blood samples were collected in tubes containing potassium EDTA (K₂EDTA) predose (prior to dosing), and at 0.5, 3, 6, 9, 12, 26, 32, 48, 72, 96, 144, 336 and 480 hours post dose and kept on ice until processed.

Blood samples were centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C, and plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C until assayed. Anti-LPA antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. At most, 25 days transpired between sample collection and analysis for all plasma samples.

3.3 Direct Binding ELISA

LT3114, LT3211, LT3212 or LT3611 antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti human IgG (gamma-chain specific) UNLB antibody (Southern Biotech 2040-01, 1 mg/mL) was diluted 1:1000 in

carbonate buffer (100mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). 384 well plates were coated with 15 µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tweenn-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting human anti-LPA antibodies in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1: 7290, 1: 21870, 1: 65610, 1: 196380 and 1: 590490 and 15µL of the standard/samples preparation were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 µL of 1: 10,000 blocking buffer diluted goat anti-human IgG HRP (Southern Biotech 2040-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15 µl/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H₂SO₄, 15 µl/well. The OD was measured at 450 nm.

3.4 Pharmacokinetic Data Analyses

Noncompartmental PK parameters were estimated using WinNonlin Pro (version 5.3; Pharsight, Mountain View, CA; Model 201) for individual animal profiles of anti-LPA plasma concentrations for the four humanized anti-LPA antibodies. Nominal times were within the time windows allowed by the protocol and were used in PK calculations. For the calculations of mean humanized anti-LPA antibody plasma concentrations, concentrations below the limit of quantification (BQL) were assigned a value as follows: 1) if all samples in a profile were BQL, a zero value for each sample was assigned, 2) at the beginning of a profile, an empty value was assigned, 3) in the middle of a profile, a BQL value was considered as an empty value, 4) at the end of a profile, the data point was considered as an empty value.

For graphical purposes, mean data are plotted as long as there are two animals that received an IV dose, donated samples, and had reported measurable assay results. For the calculations of AUC and terminal rate constant, concentrations below the limit of quantitation were assigned a value as follows: 1) at the beginning of a profile, an empty value was assigned, 2) in the middle of a profile, a BQL value was considered as an empty value, 3) at the end of a profile, the data point was considered as an empty value.

The following PK parameters were determined for 10 mg/kg IV administration of each humanized anti-LPA antibody if there were at least three measurable concentrations in each animal's profile: observed plasma concentrations at 30 minutes post IV bolus administration (C_{30min}), terminal half-life ($t_{1/2}$), and area under the plasma concentration-time curve (AUC_{last} , $AUC_{0-\infty}$). The observed last measurable concentration (C_{last}) and the time of the last measurable concentration (T_{last}) were also reported.

The slope (λ_z) of the terminal log-linear portion of each profile was determined by least squares linear regression of at least three points, and the terminal phase half-life ($t_{1/2}$) was calculated by dividing 0.693 by this slope. The linear trapezoidal method was used to determine the area under the plasma concentration-time curve from time zero to last measurable concentration (AUC_{last} , where last is the time of the last measurable concentration), or to infinity, $AUC_{0-\infty}$. Likewise the trapezoidal method was used to calculate the area under the first moment curve (AUMC; the curve of concentration*time versus time) which was subsequently used to calculate the Mean Residence Time (MRT calculated as $AUMC/AUC$).

Systemic clearance (CL) was calculated as IV dose/ $AUC_{0-\infty}$. The apparent volume of distribution of the central compartment (V_c calculated as IV Dose/ C_0), volume of distribution during the elimination phase (V_z calculated as IV Dose/ $AUC_{0-\infty} \times \lambda_z$) and volume of distribution during steady state, (V_{ss} calculated as CL x MRT) after an IV bolus dose of the humanized anti-LPA antibodies should be interpreted with the limitation that plasma data may not reflect tissue distribution if significant elimination is occurring in the peripheral or tissue compartments. Volume of distribution during steady state (V_{ss}) is reported in the tables.

PK parameters calculated to three significant figures were reported in tables to the nearest $\mu\text{g/mL}$ (C_0) or $\text{mg}\cdot\text{hr/mL}$ (AUC) after rounding of the individual parameter estimates and descriptive summary statistics. Mean humanized anti-LPA antibody plasma concentrations were calculated based on nominal times and were reported to the nearest $\mu\text{g/mL}$. Humanized anti-LPA antibody plasma concentration data and PK parameters were summarized using average values or means as well as other descriptive statistics including measures of variability (standard deviation and coefficient of variation). WinNonlin descriptive summary statistics will differ slightly from those calculated with other programs based on calculations with three significant figures.

4 RESULTS

Following a single IV dose of LT3114 to male Sprague Dawley rats, LT3114 plasma concentrations were measurable for up to 20 days post dosing (**Table 2** and **Figure 1**). LT3114 concentrations declined in a bi-exponential fashion with the initial phase occurring for up to 2 days after administration of a 10 mg/kg IV bolus. LT3114 plasma concentrations declined about three-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3114 plasma concentrations declined very slowly with an approximate terminal half-life of 3 weeks or longer (**Figure 2**). As the last sample was taken at 20 days post dosing, the estimated terminal half-life is not considered to be a reliable estimate. As a result of the unreliable estimate of terminal half-life, other related parameters such as systemic clearance and volume of distribution at steady-state are also not considered to be robust. Mean LT3114 plasma concentrations at 30 minutes were 178 $\mu\text{g/mL}$ and the mean AUC_{0-last} value was 22.9 $\text{mg}\cdot\text{h/mL}$. A significant portion of the AUC occurred beyond 20 days.

LT3211 plasma concentrations were measurable for up to 6 days post IV bolus dosing of 10 mg/kg LT3211 (**Table 3** and **Figure 1**). LT3211 concentrations declined in a bi-exponential fashion with the initial phase occurring for up to one to two days after administration of a 10 mg/kg IV bolus. LT3211 plasma concentrations declined about seven to ten-fold from 30 minutes to two days post IV bolus dosing. Thereafter, LT3211 plasma concentrations declined to non-measurable concentrations within two weeks post IV bolus dosing with an approximate terminal half-life of 5 days (**Figure 3**). Mean systemic clearance and volume of distribution at steady-state were 1.71 mL/h/kg and 203 mL/kg, respectively. Mean LT3211 plasma concentrations at 30 minutes were 181 µg/mL and the mean AUC_{0-last} value was 3.94 mg.h/mL.

For the 10 mg/kg IV bolus dose of LT3212, plasma concentrations were measurable for up to 3 days post (**Table 4** and **Figure 1**). LT3212 concentrations declined in a bi-exponential fashion with the initial phase occurring for up to one day after administration of a 10 mg/kg IV bolus. LT3212 plasma concentrations declined about five to seven-fold from 30 minutes to one day post IV bolus dosing. Thereafter, LT3212 plasma concentrations declined to non-measurable concentrations within 4 days post IV bolus dosing with an approximate terminal half-life of 1 day (**Figure 4**). Mean systemic clearance and volume of distribution at steady-state were 3.93 mL/h/kg and 102 mL/kg, respectively. Mean LT3212 plasma concentrations at 30 minutes were 145 µg/mL and the mean AUC_{0-last} value was 2.29 mg.h/mL.

Following a 10 mg/kg IV bolus dose of LT3611, plasma concentrations were measurable for up to 9 days post (**Table 5** and **Figure 1**). LT3611 concentrations declined in a bi-exponential fashion with the initial phase occurring for up to two days after administration of a 10 mg/kg IV bolus. LT3611 plasma concentrations declined about six to eight-fold from 30 minutes to two day posts IV bolus dosing. Thereafter, LT3611 plasma concentrations declined to non-measurable concentrations within 2 weeks post IV bolus dosing with an approximate terminal half-life of 3 days (**Figure 5**). Mean systemic clearance and volume of distribution at steady-state were 1.45 mL/h/kg and 94 mL/kg, respectively. Mean LT3611 plasma concentrations at 30 minutes were 194 µg/mL and the mean AUC_{0-last} value was 6.57 mg.h/mL.

Monoclonal antibody plasma concentrations were similar for all four monoclonal antibodies at 30 minutes following a 10 mg/kg IV bolus. Of the four anti-LPA monoclonal antibodies, LT3114 plasma concentrations were measurable throughout the 20 day study period. LT3114 had an estimated terminal half-life of 3 weeks or longer whereas the mean estimated terminal half-life values for the other three monoclonal antibodies (e.g., LT3211, LT3212 and LT3611) were all less than one week.

Table 2. LT3114 noncompartmental plasma pharmacokinetic parameters and plasma concentrations in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3114

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3114 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 1	Rat 2	Rat 3			
LT3114 10 mg/kg IV bolus	Predose	BLQ	BLQ	BLQ	NC	NC	NC
	0.5	203	134	198	178	39	22
	3	232	179	173	195	33	17
	6	144	121	137	134	12	9
	9	115	100	112	109	8	7
	12	123	109	143	125	17	14
	26	96	83	71	83	13	15
	32	84	95	78	86	9	10
	48	62	40	ND	51	16	31
	72	60	53	53	55	4	7
	96	52	34	44	43	9.	21
	144	48	34	39	40	7	18
	336	51	40	38	43	7	16
	480	30	25	45	33	10	31
	Noncompartmental PK Parameters				Mean ^a	SD ^a	%CV ^a
	C ₀ (µg/mL)	203	134	203	180	40	22
	T _{last} (h)	480	480	480	480	0	0
	AUC _{0-last} (mg.h/mL)	25.8	20.1	22.9	22.9	2.82	12
	AUC _{0-inf} (mg.h/mL) [AUC% extrap]	49.3 [48%]	44.2 [55%]	214 [89%]	102 [64%]	96.5 [22]	94 [35]

Table 2. LT3114 noncompartmental plasma pharmacokinetic parameters and plasma concentrations in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3114 (continued)

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3114 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 1	Rat 2	Rat 3			
	Terminal t _{1/2} (h) [days]	545 [23 days]	669 [28 days]	2940 [123 days]	1390 [58 days]	1350	97
	CL (mL/h/kg)	0.203	0.226	0.047	0.159	0.098	62
	V _{ss} (mL/kg)	144	199	198	180	32	17

BLQ=Below the limit of quantitation, NC=Not calculated, ND=Not determined, CV%=Percent coefficient of variation, SD=Standard deviation

^a At least 2 of 3 rats have measurable plasma concentrations or estimated PK parameters for calculation of descriptive statistics

Table 3. LT3211 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3211

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3211 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 4	Rat 5	Rat 6			
LT3211 10 mg/kg IV bolus	Predose	3	2	2	NC	NC	NC
	0.5	186	187	170	181	10	5
	3	201	143	128	157	39	25
	6	115	95	98	103	11	11
	9	97	98	96	97	1	1
	12	72	48	50	57	13	24
	26	37	31	25	31	6	19
	32	42	39	32	38	5	14
	48	20	17	19	19	2	8
	72	17	17	24	19	4	21
	96	13	13	15	14	1	8
	144	12	10	4*	11	1	13
	336	BLQ	4*	4*	NC	NC	NC
	480	BLQ	2*	3*	NC	NC	NC
	Noncompartmental PK Parameters				Mean ^a	SD ^a	%CV ^a
	C ₀ (µg/mL)	186	197	180	188	9	5
	T _{last} (h)	144	144	96	128	28	22
	AUC _{0-last} (mg.h/mL)	4.52	3.91	3.40	3.94	0.56	14
	AUC _{0-inf} (mg.h/mL) [AUC% extrap]	6.74 [33%]	5.57 [30%]	5.39 [37%]	5.90 [33%]	0.73 [4]	12 [11]

Table 3. LT3211 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3211 (continued)

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3211 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 4	Rat 5	Rat 6			
	Terminal t _{1/2} (h) [days]	128 [5 days]	115 [5 days]	92.1 [4 days]	112 [5 days]	18	16
	CL (mL/h/kg)	1.48	1.80	1.86	1.71	0.20	12
	V _{ss} (mL/kg)	199	217	193	203	13	6

BLQ=Below the limit of quantitation, NC=Not calculated, ND=Not determined, CV%=Percent coefficient of variation, SD=Standard deviation

^a At least 2 of 3 rats have measurable plasma concentrations or estimated PK parameters for the calculation of descriptive statistics

* Not used in calculation of descriptive statistics or PK data evaluation, concentration value not considered different than predose value

Table 4. LT3212 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3212

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3212 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 7	Rat 8	Rat 9			
LT3212 10 mg/kg IV bolus	Predose	BLQ	BLQ	BLQ	NC	NC	NC
	0.5	165	128	142	145	19	13
	3	214	113	135	154	53	34
	6	88	98	103	96	8	8
	9	44	65	74	61	15	25
	12	45	61	62	56	10	17
	26	24	24	21	23	2	8
	32	15	13	19	16	3	20
	48	7	11	10	9	2	22
	72	5	6	7	6	1	17
	96	2*	4*	3*	NC	NC	NC
	144	2*	3*	1*	NC	NC	NC
	336	BLQ	1*	BLQ	NC	NC	NC
	480	BLQ	BLQ	BLQ	NC	NC	NC
	Noncompartmental PK Parameters				Mean ^a	SD ^a	%CV ^a
	C ₀ (µg/mL)	165	131	143	147	17	12
	T _{last} (h)	72	72	72	72	0	0
	AUC _{0-last} (mg.h/mL)	2.26	2.22	2.38	2.29	0.09	4
	AUC _{0-inf} (mg.h/mL) [AUC% extrap]	2.45 [8%]	2.52 [12%]	2.67 [11%]	2.55 [10%]	0.11 [2]	4 [22]

Table 4. LT3212 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3212 (continued)

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3212 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 7	Rat 8	Rat 9			
	Terminal t _{1/2} (h) [days]	26 [1 day]	35 [1 day]	29 [1 day]	30 [1 day]	5	15
	CL (mL/h/kg)	4.08	3.97	3.74	3.93	0.17	4
	V _{ss} (mL/kg)	86	118	95	102	15	14

BLQ=Below the limit of quantitation, NC=Not calculated, ND=Not determined, CV%=Percent coefficient of variation, SD=Standard deviation

^a At least 2 of 3 rats have measurable plasma concentrations or estimated PK parameters for the calculation of descriptive statistics

* Not used in calculation of descriptive statistics or PK data evaluation

Table 5. LT3611 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3611

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3611 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 10	Rat 11	Rat 12			
LT3611 10 mg/kg IV bolus	Predose	BLQ	BLQ	BLQ	NC	NC	NC
	0.5	163	174	246	194	45	23
	3	268	182	200	217	45	21
	6	222	155	141	173	43	25
	9	134	129	139	134	5	4
	12	132	119	95	115	19	16
	26	71	47	56	58	12	21
	32	78	39	43	53	21	40
	48	27	28	31	29	2	7
	72	30	20	18	23	6	28
	96	20	11	13	15	5	32
	144	14	15	10	13	3	20
	336	5	4*	3*	NC	NC	NC
	480	3*	1*	1*	NC	NC	NC
	Noncompartmental PK Parameters				Mean ^a	SD ^a	%CV ^a
	C ₀ (µg/mL)	163	174	256	198	51	26
	T _{last} (h)	336	144	144	208	111	53
	AUC _{0-last} (mg.h/mL)	8.92	5.36	5.42	6.57	2.04	31
	AUC _{0-inf} (mg.h/mL) [AUC% extrap]	9.71 [8%]	6.15 [13%]	5.95 [9%]	7.27 [10%]	2.12 [2]	22 [25]

Table 5. LT3611 plasma concentrations and noncompartmental plasma pharmacokinetic parameters in male Sprague Dawley rats following a single IV bolus dose of 10 mg/kg LT3611 (continued)

Humanized anti-LPA mAb/ Dose/ Route	Nominal Time (hours)	Individual Male Rat LT3611 Plasma Concentration Data (µg/mL)			Mean ^a	SD ^a	%CV ^a
		Rat 10	Rat 11	Rat 12			
	Terminal t _{1/2} (h) [days]	110 [5 days]	36 [1 day]	36 [1 day]	61 [3 days]	43	70
	CL (mL/h/kg)	1.03	1.62	1.68	1.45	0.36	25
	V _{ss} (mL/kg)	107	93	82	94	13	13

BLQ=Below the limit of quantitation, NC=Not calculated, ND=Not determined, CV%=Percent coefficient of variation, SD=Standard deviation

^a At least 2 of 3 rats have measurable plasma concentrations or estimated PK parameters for the calculation of descriptive statistics

* Not used in calculation of descriptive statistics or PK data evaluation

Figure 1. Individual rat plasma concentration-time profiles after a 10 mg/kg IV bolus dose of humanized anti-LPA antibodies (Linear and Log-linear Plots)

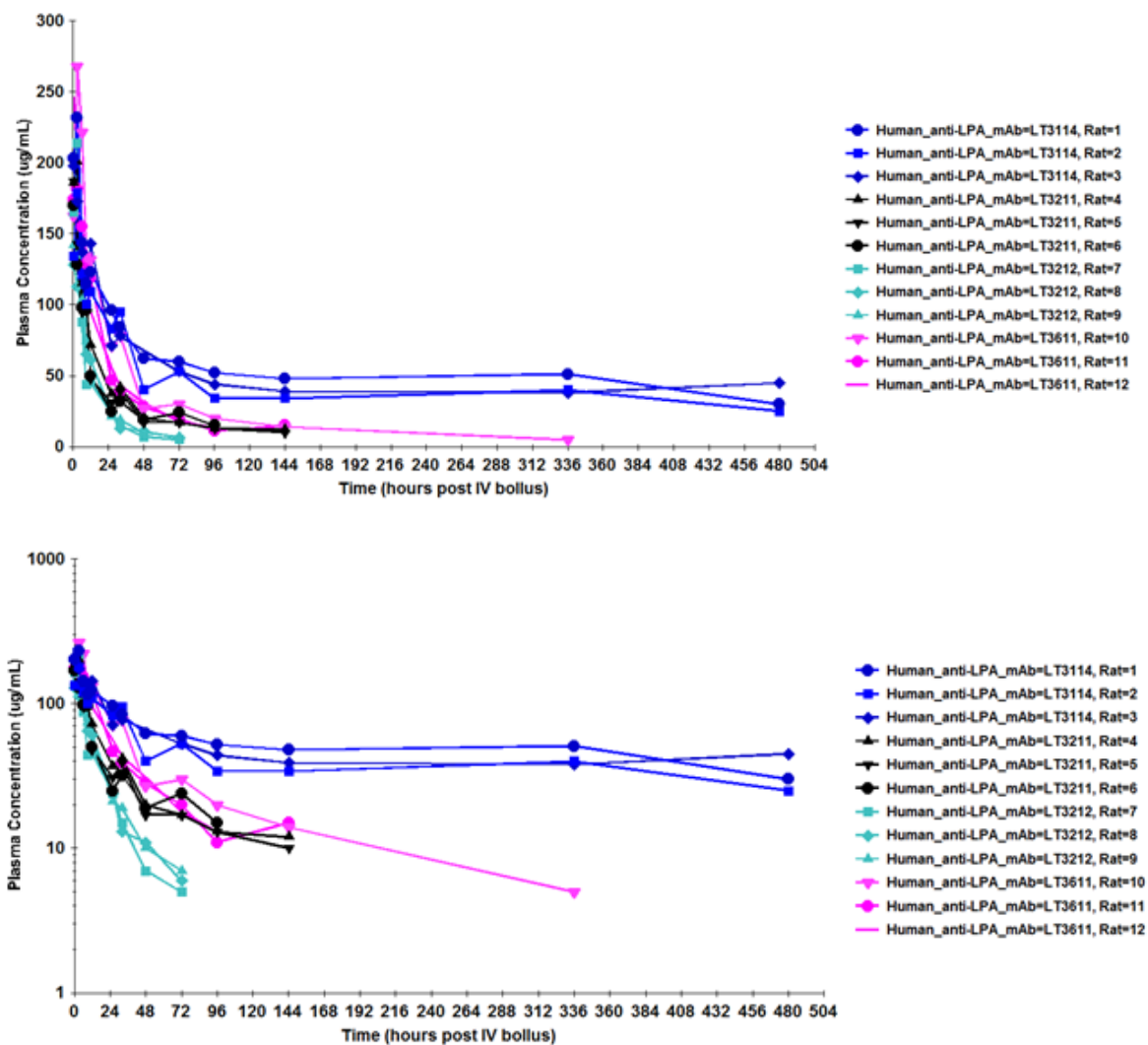


Figure 2. Terminal half-life estimates for Group 1: LT3114

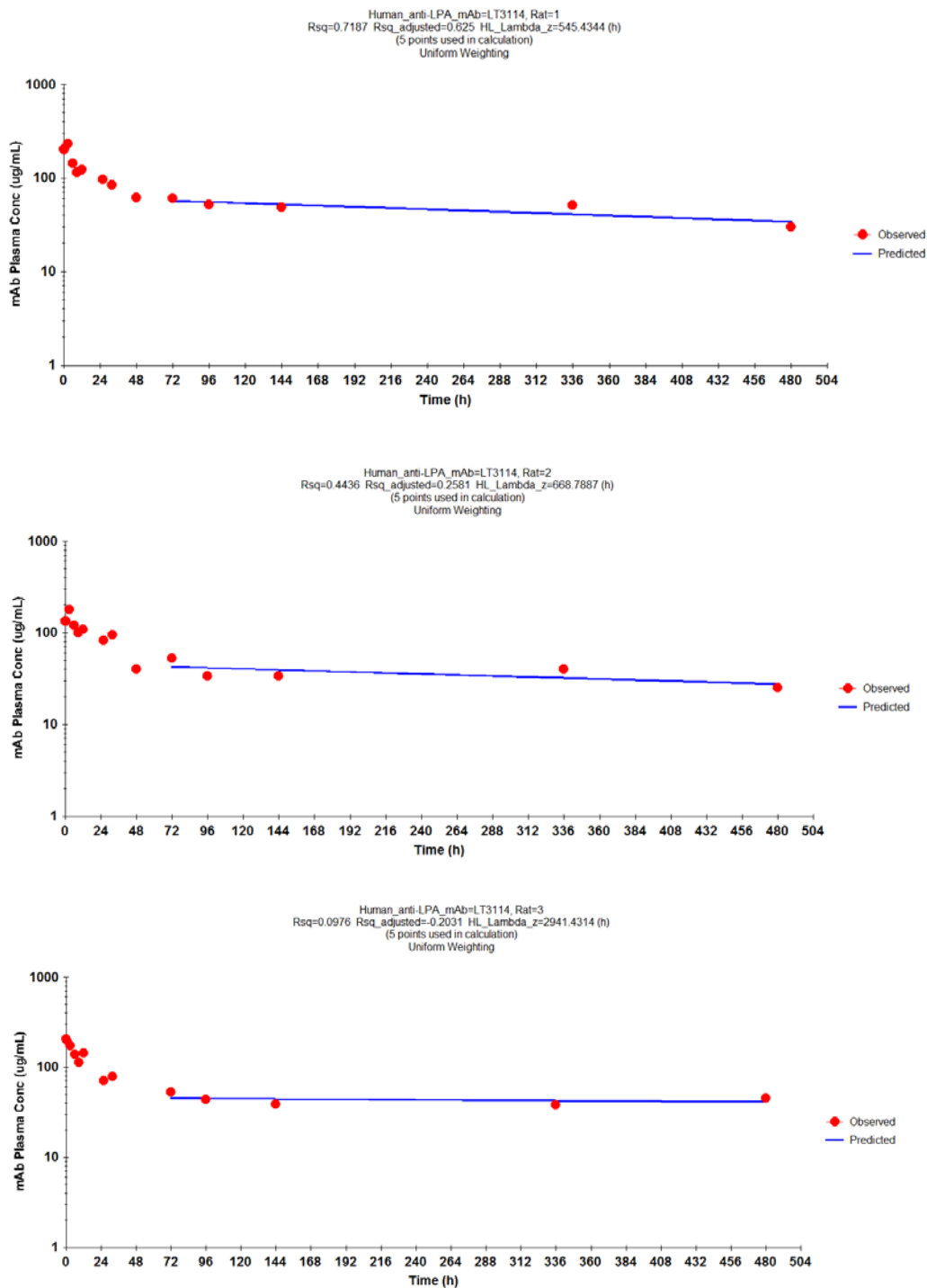


Figure 3. Terminal half-life estimates for Group 2: LT3211

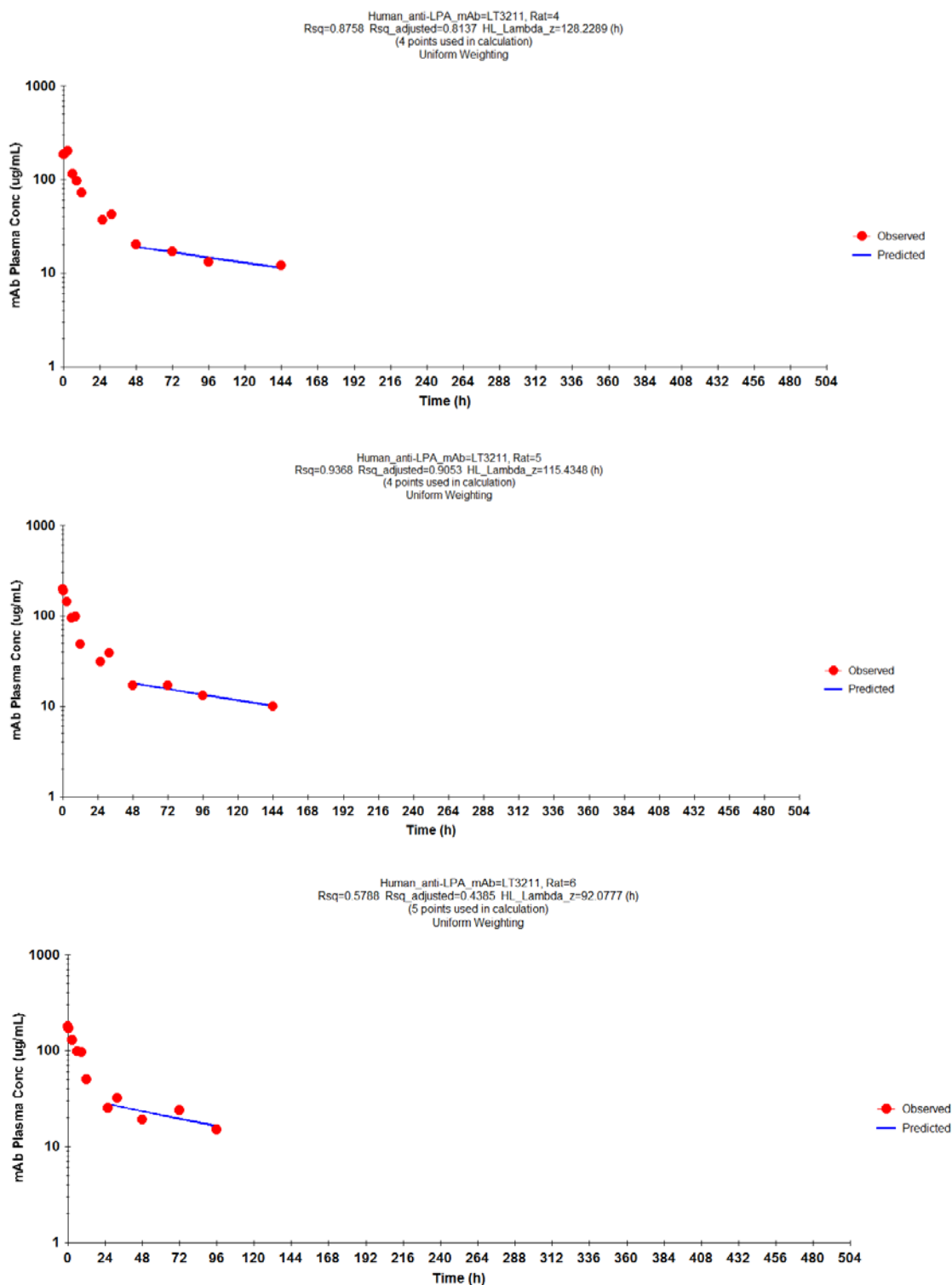


Figure 4. Terminal half-life estimates for Group 3: LT3212

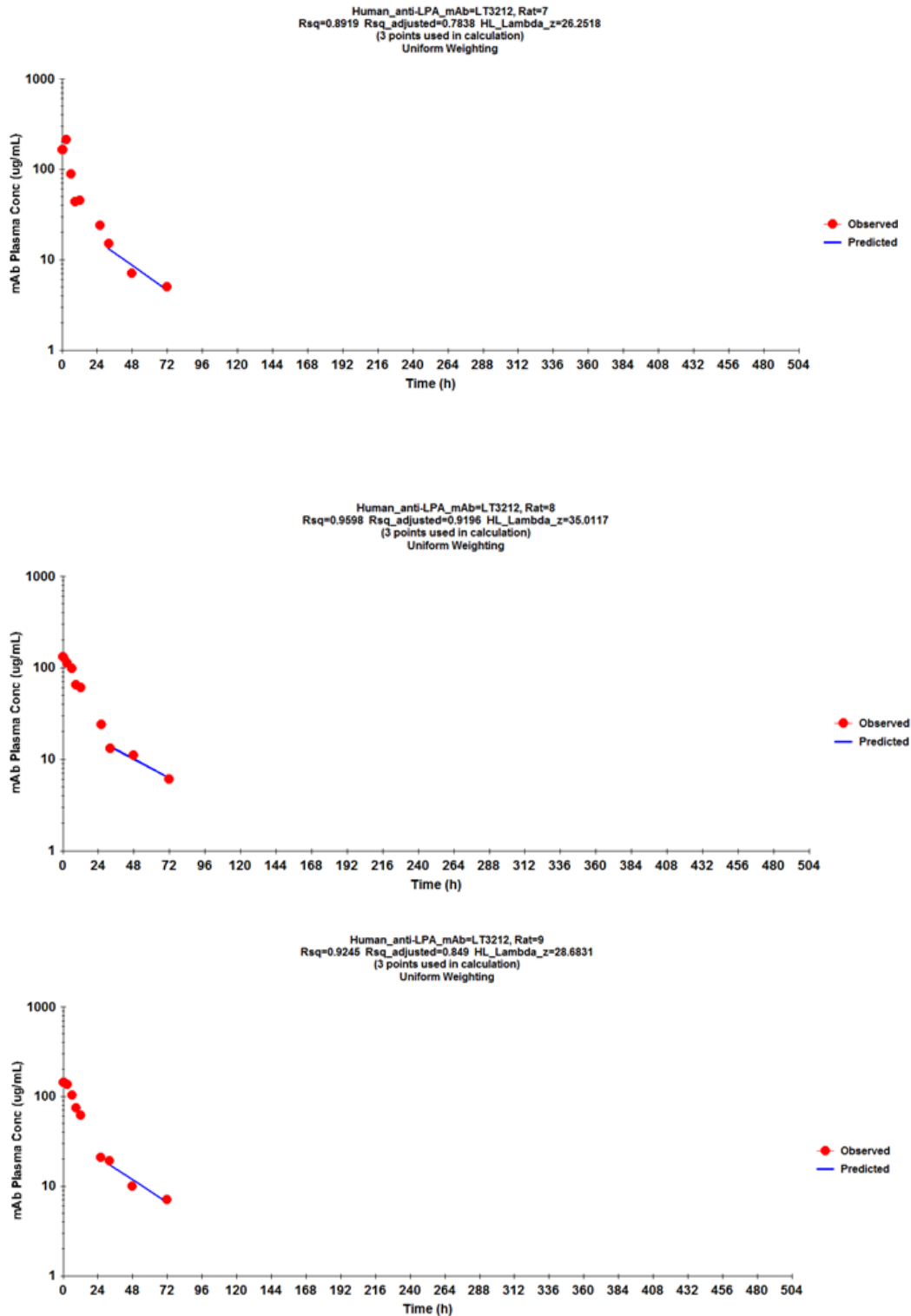
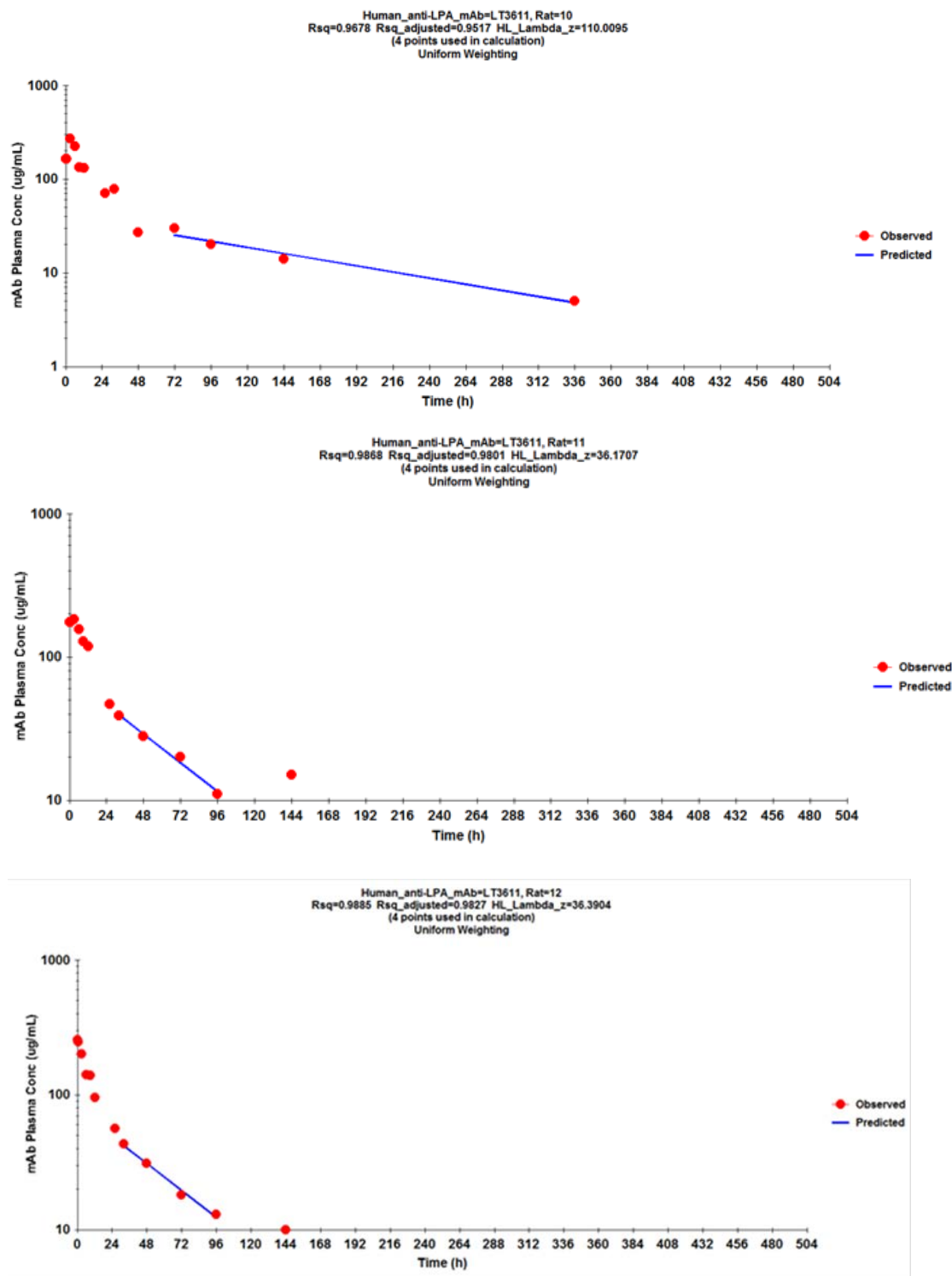


Figure 5. Terminal half-life estimates for Group 4: LT3611



5 CONCLUSION

In summary, noncompartmental PK parameters were generated for humanized anti-LPA monoclonal antibodies in Sprague Dawley rats after a single IV bolus administration of 10 mg/kg.

6 DATA ARCHIVE

Lpath Notebook # 270 pp1-14 and 28-35